

MANGAREVA, French PolynesiaWMO: **919480**Lat: **23.130S**Lon: **134.965W**Elev: **89**StdP: **100.26**Time Zone: **-9.00 (W09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.4	18.0	10.5	8.0	18.7	11.4	8.5	19.1	9.5	18.4	8.6	19.1	4.4	170	0.243

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	3.7	29.0	24.5	28.5	24.3	28.1	24.1	25.5	27.5	25.1	27.3	24.8	27.0	2.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.1	26.6	24.5	19.7	26.4	24.1	19.2	26.2	78.6	27.6	77.1	27.4	75.7	27.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.4	6.4	16.2	29.7	0.9	0.5	15.6	30.0	15.0	30.3	14.5	30.6	13.8	31.0		
(5)			12.7	25.9	0.8	0.7	12.1	26.3	11.6	26.7	11.2	27.1	10.6	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	25.5	25.9	26.0	24.8	23.4	22.2	21.5	21.2	21.4	22.2	23.4	24.6		
	DBStd	2.14	1.04	1.02	0.99	1.22	1.27	1.36	1.47	1.52	1.41	1.40	1.20	1.13		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	1	0	0	0	0	0	0	1	1	0	0	0	0		
	CDD10.0	4920	481	444	495	444	414	367	355	347	341	377	402	452		
	CDD18.3	1881	223	211	237	194	156	117	97	90	92	119	152	194		
	CDH23.3	8282	1531	1597	1784	1020	401	144	77	69	88	183	431	957		
	CDH26.7	693	161	194	212	60	5	0	0	0	1	7	53			
(14) Wind	WSAvg	2.9	2.3	2.4	2.5	2.9	2.9	3.2	3.2	2.9	3.1	3.1	3.0	2.6		
(15) Precipitation	PrecAvg	1780	161	150	153	142	148	150	143	139	115	145	170	182		
	PrecMax	2591	490	397	513	489	394	314	248	494	272	341	487	574		
	PrecMin	987	11	22	13	19	26	25	27	10	5	12	31	22		
	PrecStd	361	111	86	110	96	84	75	59	92	72	78	101	107		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.5	29.7	28.7	27.3	26.1	25.6	25.4	25.8	26.6	27.6	28.4		
		MCWB	24.4	24.8	24.8	24.6	23.6	23.0	22.3	22.2	22.1	23.0	23.5	23.9		
	2%	DB	28.6	28.9	29.0	28.0	26.5	25.4	24.8	24.7	25.1	25.7	26.7	27.8		
		MCWB	24.2	24.6	24.6	24.2	23.2	22.6	22.0	21.7	21.8	22.4	23.1	23.7		
	5%	DB	28.1	28.4	28.5	27.4	25.8	24.8	24.2	24.2	24.4	25.1	26.2	27.3		
		MCWB	24.0	24.3	24.4	23.9	22.8	22.3	21.7	21.3	21.3	22.0	22.7	23.5		
	10%	DB	27.6	27.9	27.9	26.8	25.2	24.2	23.6	23.5	23.6	24.4	25.5	26.7		
		MCWB	23.8	24.1	24.2	23.5	22.5	21.9	21.3	20.9	20.9	21.6	22.3	23.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	26.0	26.0	25.5	24.7	23.8	23.2	23.1	23.2	24.0	24.5	25.2		
		MCDB	27.4	28.0	27.9	27.4	26.0	25.1	24.7	24.5	24.7	25.3	26.2	26.9		
	2%	WB	25.2	25.3	25.4	24.9	24.1	23.2	22.7	22.5	22.6	23.3	23.9	24.7		
		MCDB	27.2	27.7	27.6	27.1	25.5	24.6	24.2	24.0	24.2	24.8	25.6	26.4		
	5%	WB	24.7	24.9	25.0	24.5	23.7	22.9	22.2	22.0	22.0	22.7	23.4	24.3		
		MCDB	26.9	27.4	27.4	26.7	25.1	24.3	23.7	23.5	23.7	24.3	25.2	26.1		
	10%	WB	24.4	24.6	24.7	24.1	23.2	22.4	21.7	21.4	21.5	22.2	23.0	23.9		
		MCDB	26.6	27.1	27.0	26.2	24.7	23.8	23.2	23.1	23.1	23.8	24.8	25.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.9	3.9	3.7	3.3	3.1	3.0	3.1	3.6	3.6	3.6	3.6	3.8		
		MCDBR	4.4	4.4	4.3	4.1	3.7	3.6	3.8	4.1	4.4	4.3	4.2	4.4		
	5% WB	MCWBR	2.3	2.3	2.1	2.1	2.2	2.3	2.5	2.5	2.6	2.4	2.3	2.2		
		MCWBR	4.1	4.1	4.0	3.8	3.4	3.3	3.6	3.9	4.1	3.9	4.0	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.366	0.369	0.362	0.366	0.358	0.351	0.346	0.347	0.358	0.359	0.362	0.369		
		taud	2.642	2.638	2.668	2.657	2.662	2.655	2.658	2.650	2.603	2.613	2.635	2.637		
	Ebn at Noon		977	961	942	897	864	850	868	903	931	960	975	976		
		Edh at Noon	100	99	92	86	79	77	78	85	96	100	100	101		
(44) All-Sky Solar Radiation	RadAvg		6.74	6.58	5.90	4.68	3.77	3.39	3.71	4.44	5.35	5.87	6.42	6.57		
	RadStd		0.82	0.57	0.41	0.46	0.28	0.25	0.18	0.23	0.38	0.50	0.56	0.59		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	-0.40	-1.06	+0.31	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page